

Revisiting Alignment for Human–AI Integration: A Proposal for Centered Evaluation

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Abstract

When humans integrate with augmentation technologies for cognitive enhancement such as brain–computer interfaces, human and AI form a single, functionally integrated system. AI alignment research, which uses human values as the reference point for designing and evaluating AI systems, faces a conceptual mismatch in these cases: the human whose values serve as the alignment target is partly constituted by the AI, while the AI that should adapt to human behavior simultaneously shapes that behavior. This paper proposes centering as a complementary concept to alignment for evaluating integrated human–AI systems. The paper outlines four dimensions of centered evaluation, i.e. subpersonal coherence, metacognitive accessibility, diachronic stability, and ecological compatibility, and motivates them based on emerging empirical evidence. The aim of this paper is not to propose centering as a replacement to alignment, but to extend it to cases where human and AI can no longer be meaningfully evaluated in isolation.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Empirical studies in HCI.

Keywords

AI alignment, human-ai integration, augmentation technologies

ACM Reference Format:

Giulia Frascaria. 2026. Revisiting Alignment for Human–AI Integration: A Proposal for Centered Evaluation. In *Proceedings of BiAlign Workshop - The ACM CHI conference on Human Factors in Computing Systems (CHI 26)*. ACM, New York, NY, USA, 3 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

AI alignment research focuses on ensuring that artificial intelligence systems act in accordance with human values, treating those values as a target against which AI behavior is measured and corrected [14]. Shen et al.'s [21] bidirectional alignment framework

advances the field by crucially recognizing mutual adaptation: human values and specifications are used to train AI systems, while humans cognitively and behaviorally adjust to AI.

However, a growing class of technologies challenges the assumption of a distinction between humans and AI. Human augmentation (HA) technologies, defined here as digital systems that extend human physical, cognitive, or sensory capabilities through computational capabilities including AI features [19], are moving from medical and niche transhumanist contexts into everyday life [12]. Brain–computer interfaces are advancing rapidly. Neuralink has begun clinical trials with a long-term vision of interfacing humans with AI directly in the brain [2]. Increasing research is exploring public perception of cognitive augmentation through AI [12], with studies indicating that 80% of respondents expect BCI and cognitive augmentation technologies to be widespread in the coming decade [8]. The global market for human augmentation reflects the increasing relevance of the field, and is projected to exceed USD 880 billion by 2032 [11]. This emerging process of cyborgization through HA technologies [1], paired with the increasing interest towards augmentation through AI features, raises a novel challenge for AI alignment research.

In this paper I propose centering as an initial concept for evaluating human-AI integrated systems, extending bidirectional alignment to cases where human and AI can no longer be meaningfully separated.

2 The Endogenous Target Problem

The concept of alignment encodes a spatial metaphor, suggesting the existence of two entities moving along the same direction. AI alignment research operationalizes this metaphor by treating human values as an exogenous target, i.e. a reference point independent of the AI system being evaluated [21]. However, integration of humans with AI through augmentation technologies destabilizes this assumption by making the target endogenous. "Aligning" the AI to those values means aligning it to a target that is partially produced by the system itself. This problem is acknowledged in existing HCI research. Frauenberger [13] characterizes the increasing human–technology entanglement as a continuous coupling, leading to the dissolution of ontological and epistemological boundaries between the human and the technology. Under the assumption of an entanglement between the human and the AI, such as the one that is resulting from human-AI integration through augmentation technologies, asking whether a system aligns with a person's values presupposes a distinction that is instead dissolved.

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CHI 26, Barcelona, Spain

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ACM ISBN 978-1-4503-XXXX-X/2026/04
<https://doi.org/XXXXXXX.XXXXXXX>

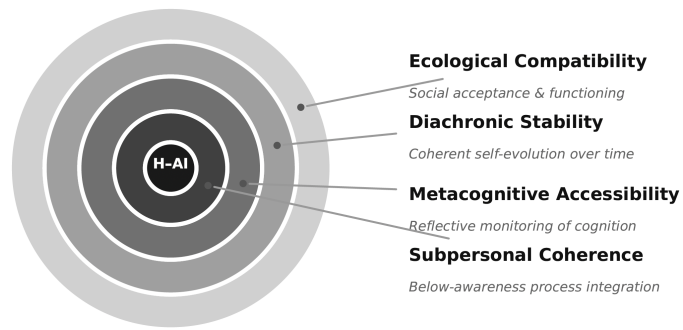


Figure 1: Concentric dimensions of centered evaluation. Inner rings represent individual-level processes; outer rings extend to temporal and social dimensions.

3 Toward a Centered Evaluation

This paper proposes a framework for a centered evaluation of integrated human-AI systems. Where AI alignment research asks whether the two entities move in the same direction, centering asks whether the merger of human and AI creates a single, coherent, stable entity. The following paragraphs present four concentric dimensions for a centered evaluation. Each dimension is based on emerging empirical findings and theoretical frameworks. Figure 1 illustrates the four concentric dimensions. Where alignment asks whether two entities move in the same direction, centering asks whether their merger produces a single coherent entity, evaluated from the inside out.

Inner ring: Subpersonal centering. The first dimension evaluates whether AI-augmented and non-augmented cognitive processes cohere below the level of conscious awareness. This draws on the personal/ subpersonal distinction in cognitive science [6, 7], which separates processes accessible to the person (beliefs, decisions, reflections) from those that operate beneath awareness (neural firing, perceptual processing, automatic responses). In a human-AI integrated system, AI can affect both levels.

Villa et al. [22] provide an empirical illustration of the open research problem. In their study, thirty participants performed a Columbia Card Task, a gambling game where participants flip cards to accumulate wins while risking losses. Participants were told they were being augmented with a brain-computer interface that played inaudible sounds to enhance cognitive function. In reality, no sounds were played. Despite the augmentation being entirely fictitious, participants in the placebo condition showed increased risk-taking behavior. Critically, their electroencephalogram (EEG) data revealed altered event-related potentials when flipping loss cards under placebo augmentation, a neural response that differed from the control condition. If the integration with cognitive augmentation produces conflicts at a subpersonal level, alignment’s reliance on expressed values as a reference point is undermined. Consequently, this finding motivates subpersonal coherence as a necessary evaluative dimension for human-AI integrated systems.

Second ring: Metacognitive accessibility. The second dimension evaluates whether the person can reflectively monitor their augmented cognition and experience it as their own. This draws on

the concept of metacognition, i.e. the awareness and regulation of one’s own cognitive processes [10], and on self-determination theory [5], which describes how people regulate external influences along a continuum from introjection, where a process is adopted but remains felt as external pressure, to internalization, where it becomes experienced as genuinely one’s own.

Recent empirical work underscores this dimension’s relevance. Fernandes et al. [9] conducted two large-scale studies demonstrating that AI use impairs metacognitive monitoring. Participants who used a large language model to solve logical reasoning tasks showed improved performance but significantly overestimated their abilities, and higher AI literacy correlated with lower metacognitive accuracy. The classic Dunning–Kruger effect ceased to exist under AI use conditions, replaced by uniform overconfidence across all skill levels. While the connection between this effect and cognitive augmentation is not established, the increased risk-taking behavior highlighted by Villa et al. [22] is indicative that this dimension should be further explored to evaluate human-AI integrated systems.

Welsch [24] proposes a research program for measuring metacognitive accuracy and neural signatures during AI interaction through BCIs, which would be essential for operationalizing this dimension.

Third Ring: Diachronic stability. Diachronic stability concerns whether the effects of human-AI integration unfold in a manner that preserves coherent evolution of the person over time, rather than producing uncontrolled or irreversible transformation. This dimension draws on James’ [16] conceptualization of the self as continuity of consciousness across a series of experiences over time. In the context of human-AI integration, diachronic stability requires that the integrated system does not evolve uncontrollably due to opaque shifts driven by the technology. Cognitive augmentation that improves present performance may erode unaided capacities over time as exemplified by research on cognitive offloading [20], which shows that reliance on external cognitive tools can atrophy internal capacities. A clinical case from a first-in-human BCI trial illustrates the relevance of this concern: after BCI implantation for seizure prediction, a patient reported increased agential capacity and continuity, but following device explantation experienced persistent agential discontinuity and traumatic harm [15]. Consequently, aligning with More’s [18] conceptualization of the diachronic self, the dimension of diachronic stability foregrounds the value of self-directed change, highlighting that augmentation-induced transformations should trace back to the person’s own reflective endorsements. Key aspects of this dimension may thus include reversibility, the preservation of core capacities in the absence of augmentation, and whether the person can later recognize and affirm the role the system played in shaping them.

Fourth ring: Ecological compatibility. This dimension evaluates whether the human-AI integrated system functions within its social, professional, and institutional environment. It emerges from Villa et al.’s [23] work on society’s attitudes towards human augmentation, which identifies social threat and perceived agency as the two primary constructs shaping societal acceptance of augmented individuals. Their findings show that an augmented person can be perceived as having low agency over their augmentation, as well as a threat towards its surroundings. This suggests that ecological

centering requires not only that the integrated entity functions well in its environment, but that it is integrated in its social context.

4 Discussion and Implications

This short article introduced the concept of *centering* to expand the research field of human–AI alignment towards human–AI integrated systems. The centered evaluation is organized as four concentric dimensions. The four dimensions are exploratory and invite discussion on their adequacy, but highlight the interdisciplinary nature of the field, which also implies a diversity of research methods that are suitable for each of the dimensions.

While this paper has primarily drawn on brain–computer interfaces to motivate the centered evaluation, the endogenous target problem is not limited to physical augmentation. AI companion technologies that function as conversational AI systems designed for sustained emotional engagement, present a structurally analogous challenge. Kirk et al. [17] characterize this as a problem of socioaffective alignment, noting that the human–AI relationship shapes the user’s preferences and perceptions through mutual influence. Empirical evidence supports the relevance of the centered dimensions in this context. At the metacognitive level, users of AI companions exhibit impaired self-monitoring; they overestimate their own capacities [9] and develop emotional dependencies they may not fully recognize [4]. Diachronic stability is directly implicated by De Freitas et al.’s [3] finding that when an AI companion’s features are altered, users experience identity discontinuity and grief-like responses, suggesting that the person’s sense of self had become entangled with the system. At the ecological level, concerns have been raised that sustained AI companion use can displace human social bonds and distort relational expectations [4]. These parallels suggest that centering may offer evaluative resources beyond the BCI context, extending to any case where prolonged human–AI interaction produces functional integration, even without a physical interface.

A limitation of the present proposal is that the four rings remain largely theoretical. While motivated by early empirical evidence, their conceptual stabilization and operationalization as evaluative standards for human–AI integrated systems requires substantial further work. The present paper hopes to be a useful starting point for future discussions and research.

Acknowledgments

We thank the University of Zurich and the Digital Society Initiative for financing this research.

References

- [1] Woodrow Barfield and Alexander Williams. 2017. Cyborgs and enhancement technology. *Philosophies* 2, 1 (2017), 4.
- [2] O. Rose Broderick. 2026. Neuralink’s big vision collides with reality of brain implants. *STAT* (January 5 2026). <https://www.statnews.com/2026/01/05/neuralink-brain-computer-interface-medical-device-vs-transhumanism/> Accessed: 2026-02-17.
- [3] Julian De Freitas, Noah Castelo, Ahmet K Uğuralp, and Zeliha Oğuz-Uğuralp. 2024. Lessons from an app update at Replika AI: identity discontinuity in human–AI relationships. *arXiv preprint arXiv:2412.14190* (2024).
- [4] Julian De Freitas and I Glenn Cohen. 2025. Unregulated emotional risks of AI wellness apps. *Nature Machine Intelligence* 7, 6 (2025), 813–815.
- [5] Edward L Deci and Richard M Ryan. 2012. Self-determination theory. *Handbook of theories of social psychology* 1, 20 (2012), 416–436.
- [6] Daniel C Dennett. 2002. *Content and consciousness*. Routledge.
- [7] Zoe Drayson. 2014. The personal/subpersonal distinction. *Philosophy compass* 9, 5 (2014), 338–346.
- [8] Austen El-Osta, Mahmoud Al Ammouri, Shujhat Khan, Sami Altalib, Manisha Karki, Eva Riboli-Sasco, and Azeem Majeed. 2025. Community perspectives regarding brain–computer interfaces: A cross-sectional study of community-dwelling adults in the UK. *PLOS Digital Health* 4, 2 (2025), e0000524.
- [9] Daniela Fernandes, Steeven Villa, Salla Nicholls, Otso Haavisto, Daniel Buschek, Albrecht Schmidt, Thomas Kosch, Chenxinran Shen, and Robin Welsch. 2025. AI makes you smarter but none the wiser: The disconnect between performance and metacognition. *Computers in Human Behavior* (2025), 108779.
- [10] John H Flavell. 1979. Metacognition and cognitive monitoring: A new area of cognitive–developmental inquiry. *American psychologist* 34, 10 (1979), 906.
- [11] Fortune Business Insights. 2025. Human Augmentation Market Size, Share & Trends, 2024–2032. *Fortune Business Insights*. <https://www.fortunebusinessinsights.com/human-augmentation-market-107046> Accessed: 2025-09-10.
- [12] Giulia Frascaria, Daniela Jaramillo-Dent, and Michael Latzer. 2026. Adoption of Human Augmentation Technologies for Non-Medical Applications: A Systematic Review of Empirical Literature. *International Journal of Human–Computer Interaction* 0, 0 (2026), 1–33. doi:10.1080/10447318.2026.2625258 arXiv:https://doi.org/10.1080/10447318.2026.2625258
- [13] Christopher Frauenberger. 2019. Entanglement HCI the next wave? *ACM Transactions on Computer-Human Interaction (TOCHI)* 27, 1 (2019), 1–27.
- [14] Iason Gabriel. 2020. Artificial intelligence, values, and alignment. *Minds and machines* 30, 3 (2020), 411–437.
- [15] Frederic Gilbert, Marcello Ienca, and Mark Cook. 2023. How I became myself after merging with a computer: Does human–machine symbiosis raise human rights issues? *Brain Stimulation* 16, 3 (2023), 783–789.
- [16] William James, Frederick Burkhardt, Fredson Bowers, and Kęstutis Skrupskelis. 1890. *The principles of psychology*. Vol. 1. Macmillan London.
- [17] Hannah Rose Kirk, Iason Gabriel, Chris Summerfield, Bertie Vidgen, and Scott A Hale. 2025. Why human–AI relationships need socioaffective alignment. *Humanities and Social Sciences Communications* 12, 1 (2025), 1–9.
- [18] Max More. 1995. *The diachronic self: identity, continuity, transformation*. University of Southern California.
- [19] Roope Raisamo, Ismo Rakkolainen, Päivi Majaranta, Katri Salminen, Jussi Rantala, and Ahmed Farooq. 2019. Human augmentation: Past, present and future. *International Journal of Human-Computer Studies* 131 (2019), 131–143.
- [20] Evan F Risko and Sam J Gilbert. 2016. Cognitive offloading. *Trends in cognitive sciences* 20, 9 (2016), 676–688.
- [21] Hua Shen, Tiffany Kneare, Reshmi Ghosh, Kenan Alkiek, Kundan Krishna, Yachuan Liu, Savvas Petridis, Yi-Hao Peng, Li Qiwei, Chenglei Si, et al. 2024. Position: Towards Bidirectional Human–AI Alignment. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems Position Paper Track*.
- [22] Steeven Villa, Thomas Kosch, Felix Grelka, Albrecht Schmidt, and Robin Welsch. 2023. The placebo effect of human augmentation: Anticipating cognitive augmentation increases risk-taking behavior. *Computers in Human Behavior* 146 (2023), 107787.
- [23] Steeven Villa, Jasmin Niess, Albrecht Schmidt, and Robin Welsch. 2023. Society’s attitudes towards human augmentation and performance enhancement technologies (SHAPE) scale. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 3 (2023), 1–23.
- [24] Robin Welsch. [n.d.]. Amplifying the mind with interactive AI. ([n.d.]).

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009